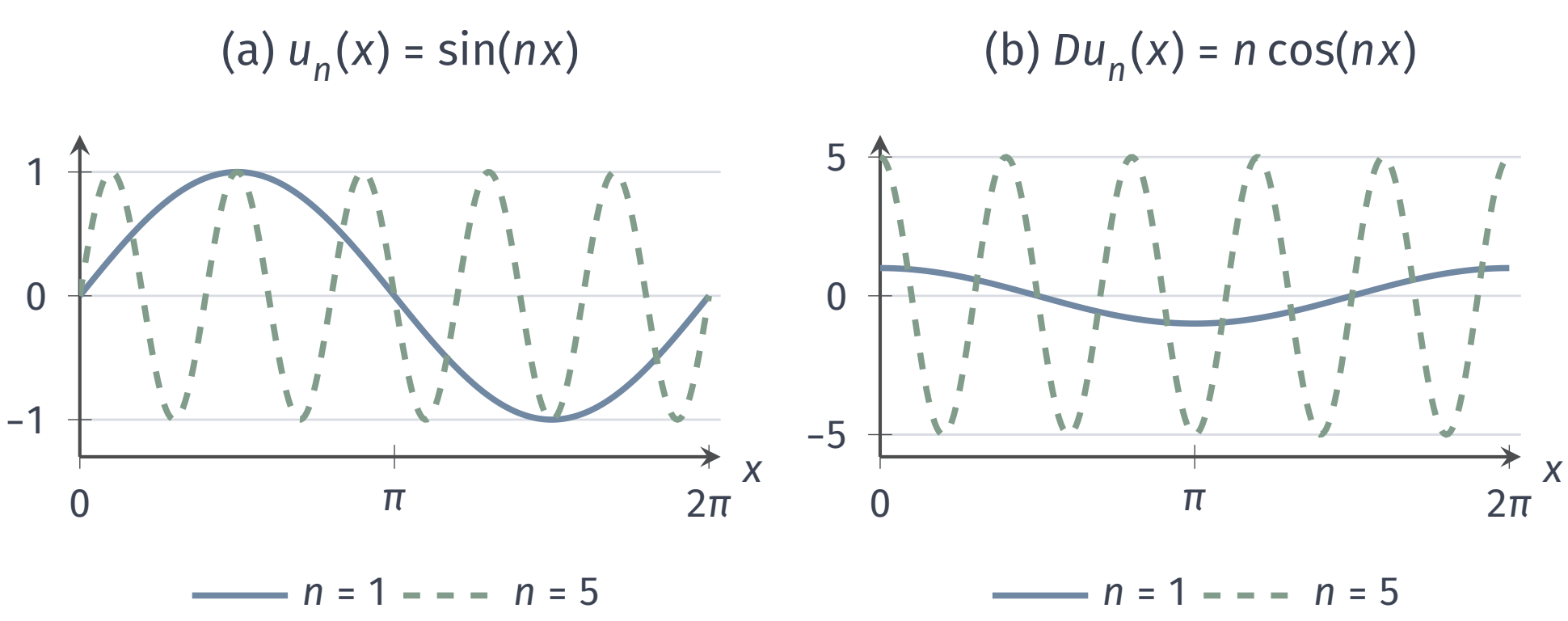
(a) $u_n(x) = \sin(nx)$
1
0
−1
0
π
2π
x
n = 1
n = 5
(b) $Du_n(x) = n\cos(nx)$
5
0
−5
0
π
2π
x
n = 1
n = 5